Module 56 - DOM Traversing

1. What is DOM Traversing?

DOM Traversing is the process of moving between related HTML elements in the DOM tree, such as parent, child, and sibling elements.

DOM Traversing Properties

I. Parent Properties

- **parentElement** → Returns the parent element.
- **parentNode** → Returns the parent node.

II. Child Properties

- **childElementCount** → Returns the number of child elements.
- **childNodes** → Returns all child nodes, including text nodes.
- **children** → Returns all child elements.
- **firstChild** → Returns the first child node.
- **firstElementChild** → Returns the first child element.
- **lastChild** → Returns the last child node.
- **lastElementChild** → Returns the last child element.

III. Sibling Properties

- **nextSibling** → Returns the next sibling node.
- **nextElementSibling** → Returns the next sibling element.
- **previousSibling** → Returns the previous sibling node.
- **previousElementSibling** → Returns the previous sibling element.

Important Difference: Node properties can include text nodes, while Element properties consider only HTML elements.

Example Code:

Index.html

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>DOM Traversing</title>
</head>
<body>
    <!-- Parent element containing child elements and text nodes -->
    <div class="parent">
        Javascript
        <div class="child1">Content 1</div>
        Dynamically Typed
        <div class="child2">Content 2</div>
        Programming Language
        <div class="child3">Content 3</div>
        Single Threaded
    </div>
    <!-- Calls the function when the button is clicked -->
    <button onclick="traversingParent()">Parent Properties</button>
    <!-- Calls the function when the button is clicked -->
    <button onclick="selectChild()">Child Properties</button>
    <!-- Calls the function when the button is double-clicked -->
    <button ondblclick="selectSiblings()">Double Click for Siblings Properties</button>
    <script src="index.js"></script>
</body>
</html>
```

Index.js

```
// Parent Properties
function traversingParent(){
    let getParent = document.querySelector(".child1");
    console.log(getParent.parentElement);
    console.log(getParent.parentNode);
    let getParent1 = document.querySelector("html");
    console.log(getParent1.parentElement);
    console.log(getParent1.parentNode);
}

// Node ---> Element Node, Text Node, Attribute Node, Document (Root Node)

// Child Properties
function selectChild(){
    let getElementByClass = document.querySelector(".parent");
    console.log(getElementByClass.childElementCount);
    console.log(getElementByClass.childNodes);
    console.log(getElementByClass.children);
    console.log(getElementByClass.firstChild);
    console.log(getElementByClass.firstElementChild);
    console.log(getElementByClass.lastChild);
    console.log(getElementByClass.lastElementChild);
}

// Sibling Properties
function selectSiblings(){
    let child1 = document.querySelector(".child1");
    console.log(child1.nextSibling);
    console.log(child1.nextElementSibling);
    console.log(child1.previousSibling);
    console.log(child1.previousElementSibling);
}
```